Dynamical Neural Networks: System Theory, Learning, and Computational Complexity

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Outline

- Dynamic (recurrent) nets
- System theory
  - observability
  - identifiability
  - minimality
  - reachability
  - approximation
- Learning behavior from i/o data
- Universal model of analog computing

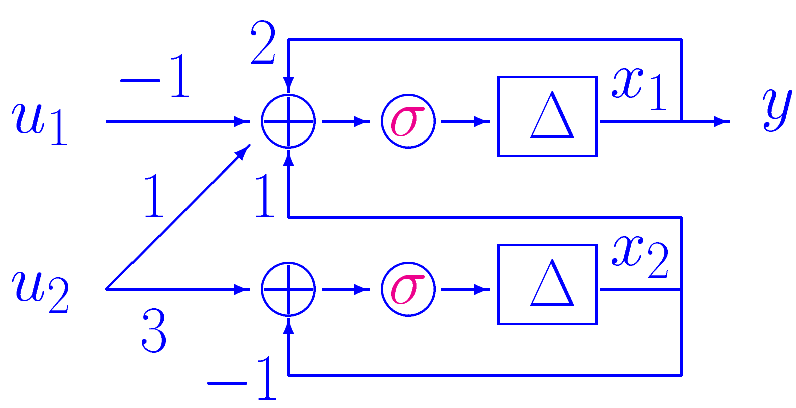
Recurrent or Dynamic Nets

Example (discrete time, two-dimensional):

$$\begin{aligned} x_1(t+1) &= \sigma(2x_1(t) + x_2(t) - u_1(t) + u_2(t)) \\ x_2(t+1) &= \sigma(-x_2(t) + 3u_2(t)) \\ y(t) &= x_1(t) \end{aligned}$$

$\sigma : \mathbb{R} \to \mathbb{R}$ "activation" (saturation)

x_i internal state of ith processor ("neuron")
$(u_1(t), u_2(t))$ external input
$y(t)$ measured response (output)



Or, differential equations $\dfrac{dx_i}{dt} = \sigma(\ldots)$

General Model

Fix $\sigma : \mathbb{R} \to \mathbb{R}$ Lipschitz, odd (for simplicity of expo)

$$\vec{\boldsymbol{\sigma}} \, (= \vec{\boldsymbol{\sigma}}_n) \; : \; \mathbb{R}^n \to \mathbb{R}^n \; : \; \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \mapsto \begin{pmatrix} \sigma(x_1) \\ \vdots \\ \sigma(x_n) \end{pmatrix}$$

n-dimensional, m-input, p-output *recurrent net :*

$$\begin{aligned} \dot{x}(t) \; [\text{or } x(t+1)\,] \; &= \; \vec{\boldsymbol{\sigma}} \, (Ax(t) + Bu(t)) \\ y(t) \; &= \; Cx(t) \end{aligned}$$

specified by: $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$, $C \in \mathbb{R}^{p \times n}$
(weights, intensities, synaptic strengths)

e.g.: $A = \begin{pmatrix} 2 & 1 \\ 0 & -1 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 1 \\ 0 & 3 \end{pmatrix}$, $C = (1 \quad 0)$

more generally, coords of $y(t)$ = outputs of p probes
each averages activation values of several "neurons"

write $\Sigma = (A, B, C, \sigma)$

Model Appears in Much Current Work

- associative memory and optimization (Hopfield)
- control of robotic manipulators (Jordan)
- speech recognition (Fallside, Kuhn)
- speaker identification (Anderson)
- formal language inference (Giles)
- sequence extrapolation for time series prediction (Farmer)
- signal processing (Bengio's book)
- adaptive control (Narendra)
- special purpose chips
- complicated dynamics
- but nice algebraic structure - generalize linear ($\sigma = \text{id}$)
- nice theory!

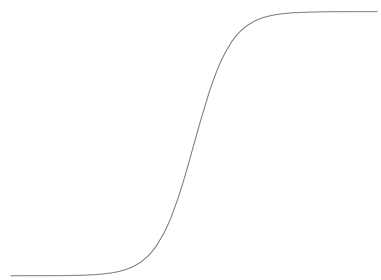
sometimes variations such as

$$\dot{x} \;=\; -Dx + \vec{\boldsymbol{\sigma}}\,(Ax + Bu)\,, \quad y = Cx\,,$$

— for many problems, may reduce study to $D = 0$

Possible Properties of σ

all will be OK for
$\sigma(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$
i.e., up to translations,
"logistic" $1/(1+e^{-x})$



more generally, various properties used:

σ satisfies IP (independence property) if
translates/dilations $1, \sigma(a_1 s + b_1), \ldots, \sigma(a_l s + b_l)$
linearly independent, $\forall$ distinct (a_i, b_i), $a_i > 0$

σ satisfies SP (spanning property) if
translates/dilations $1, \sigma(a_1 s + b_1), \ldots, \sigma(a_l s + b_l)$
span C^0, $\forall$ distinct (a_i, b_i), $a_i > 0$ (closure; on compacts)

IP: OK tanh, arctan, ... suft extends to analytic on strip
$S = \{|\mathrm{Im} z| \leq d\} \setminus \{z_0, \bar{z}_0\}$, ($z_0$ singularity, $\mathrm{Im} z_0 = d$)
(Albertini-EDS'92) – also asympt suft conds

SP: OK $\forall$ loc Riemann integrable non-polynomial
(Leshno/Pinkus'93) (if not odd, ask all a_i)

Some Notations

$\xi \in \mathbb{R}^n$, $u : [0, T] \to \mathbb{R}^m$ (measurable ess bd)

$x_{\xi,u}(\cdot)$ = soln of IVP :

$$\dot{x} = \vec{\boldsymbol{\sigma}}(Ax + Bu), \quad x(0) = \xi$$

similarly for discrete time

Let $\mathcal{B}$:= class of all $B \in \mathbb{R}^{n \times m}$ s.t.:

- $(\forall\, i)$ $\mathrm{row}_i(B) \neq 0$
- $(\forall\, i \neq j)$ $\mathrm{row}_i(B) \neq \pm \mathrm{row}_j(B)$

($m = 1$: for vector b entries nonzero, $\neq$ magnitudes)
complement of class is algebraic subset of $\mathbb{R}^{n \times m}$
so set generic (full measure, open dense)

Internal State Diagnosis (Observability)

can infer activation values from i/o data $(u(\cdot), y(\cdot))$?

observability :

$$C\,x_{\xi,u} \;\equiv\; C\,x_{\zeta,u}\;\;\forall\, u(\cdot) \quad\Rightarrow\quad \xi = \zeta$$

graph G: nodes $\{1,\ldots,n\}$, edge $i \rightsquigarrow j$ if $a_{ij} \neq 0$
G_0:= nodes i s.t. ith column of C is $\neq 0$
"every variable influences the output":
if every node reachable from G_0

Theorem. (Albertini+EDS'94) (d.t. & c.t., $B\in\mathcal{B}$, $\sigma\in$IP)

Observable iff e.v.i.o. and rank $\begin{pmatrix} A \\ C \end{pmatrix} = n$

necessity: obvious e.v.i.o. and
$C\xi = 0 \Rightarrow$ same instantaneous measurements as for $x(0)$=0
$A\xi = 0 \Rightarrow \vec{\boldsymbol{\sigma}}(A\xi + Bu) = \vec{\boldsymbol{\sigma}}(A0 + Bu)$ so same later too
sufficiency takes work: coord-invariant subspaces in ker C

Form Determined by Function

realization problem (Springer L.N.'79):
suppose we have model of the I/O operator

$$u(\cdot) \longrightarrow \boxed{\Sigma, \xi\ ?} \longrightarrow y(\cdot)$$

are the weights/parameters uniquely determined?
Answer: yes! (up to state exchanges and/or sign flips)

denote $\lambda_{\Sigma,\xi} : u(\cdot) \mapsto y(\cdot) = Cx_{\xi,u}(\cdot)$ i/o map
i/o equivalent if: $m_1{=}m_2$, $p_1{=}p_2$, $\lambda_{\Sigma_1,\xi_1}{=}\lambda_{\Sigma_2,\xi_2}$

sign-permutation (s/p) equivalent: $n_1 = n_2$ and
$\exists\, T = PQ$, $P =$ permutation, $Q = \mathrm{Diag}\,(\pm 1, \ldots, \pm 1)$
s.t. $TA_2 = A_1T$, $C_2 = C_1T$, $TB_2 = B_1$, $Tx_2 = x_1$

Theorem. (Albertini-EDS'92,. . .)
($\sigma \in \mathrm{IP}$, Σ_i's observable, $B_i \in \mathcal{B}$, discr or cont time)

$$(\Sigma_1, \xi_1) \underset{\mathrm{i/o}}{\sim} (\Sigma_2, \xi_2) \iff \textbf{s/p equivalent}$$

Model Minimality

assume same operator also realized by (Γ, ζ):

$$\dot{z} = f(z,u)\,,\ z(0) = \zeta\,, \quad y = h(z)$$

states $\in Z$ = paracompact (real-)analytic conn manifold
$h : Z \to \mathbb{R}^p$ analytic
$f : Z \times \mathbb{R}^m \to TZ$ continuous, $\pi(f(z,u)) = z$
$f(z,u)$ analytic on z, f_z continuous on $Z \times \mathbb{R}^m$
and well-posed

"i/o equivalent" if outputs coincide for all possible inputs

Theorem. (for (Σ, ξ), σ=tanh, $B \in \mathcal{B}$, observable)

$$(\Gamma, \zeta) \underset{\mathrm{i/o}}{\sim} (\Sigma, \xi) \Rightarrow \dim Z \geq \dim \Sigma$$

moreover: $\exists$ analytic submanifold Z_0 of Z
and analytic onto $\Pi : Z_0 \to \mathbb{R}^n$ s.t.
$h(q) = C\Pi(q)\ \forall\, q \in Z_0$
and $\forall\, u : [0,T] \to \mathbb{R}^m$, $t \in [0,T]$,
$z_{\zeta,u}(t) \in Z_0$, $\Pi(z_{\zeta,u}(t)) = x_{\xi,u}(t)$

Possible State Configurations?

$\bigcup_{T\geq 0}\mathcal{L}_\infty^m[0,T]$ (concatenation semigroup) acts on $\mathbb{R}^n$
via soln of IVP : $(\xi,u)\mapsto x_{\xi,u}(T)$

When transitive?

Theorem. (Sussmann+EDS'96) ($\sigma = \tanh$; cont time)

Transitive ($\forall\, A\in\mathbb{R}^{n\times n}$) iff $B\in\mathcal{B}$

uses $\lim_{s\to+\infty}\frac{\sigma(+\infty)-\sigma(as+b)}{\sigma(+\infty)-\sigma(s)} = 0 \quad \forall\, b, \forall\, a>1$

$\rightsquigarrow\; T_\xi\mathbb{R}^n = \text{conv cone}\,\{\vec{\boldsymbol{\sigma}}(A\xi+Bu), u\in\mathbb{R}^m\}$

false for other IP; e.g. not transitive:

$$
\begin{aligned}
\dot{x}_1 &= \arctan(x_1+x_2+x_3+x_4+2u)\\
\dot{x}_2 &= \arctan(x_1+x_2+x_3+x_4+12u)\\
\dot{x}_3 &= \arctan(-3u)\\
\dot{x}_4 &= \arctan(-4u)
\end{aligned}
$$

note: open problem for discrete time, on (range of σ)^n

Approximation

given i/o behavior $u(\cdot) \mapsto y(\cdot)$ realized by (Γ, ζ):

$$\dot{z} = f(z,u)\,,\;\; z(0) = \zeta\,, \quad y = h(z)$$

(assume $z(t) \in \mathbb{R}^q$, well-posed)

$$u(\cdot) \quad \longrightarrow \boxed{\quad \Gamma, \zeta \quad} \longrightarrow \quad y(\cdot)$$

Remark (EDS'92): using nets with $\sigma \in$ SP
can approximate arbitrarily close
on equicontinuous sets of inputs, finite intervals

skipping (routine) statement of approx theo
need dim $n \to \infty$ for better approx, of course

based on density of "1HL nets" to approx RHS
(Cybenko, Hornik-White-etc...)
plus extra work to eliminate "hidden layers"

stability ("fading memory") $\rightsquigarrow T = \infty$
analog also valid in d.t.

Noisy Identification / PAC Learning

cf. Koiran-EDS NIPS'95 & EuroCOLT'97;
Dasgupta-EDS IEEE-IT'96; EDS'97

$$u(\cdot) \longrightarrow \boxed{\quad ? \quad} \longrightarrow y(\cdot)$$

random inputs $u_1(\cdot), \ldots, u_s(\cdot)$; ensuing $y_1(\cdot), \ldots, y_s(\cdot)$
suppose Σ_{best} = net (Σ, ξ) that best fits data

is Σ_{best} a good model?

- 'random' = ?
- 'best fit' = ?
- 'good model' = ?
- s ('sample complexity') = ?

Vapnik/Pollard/Haussler[/Valiant] technical framework
≈ uniform convergence empirical means (∼Glivenko-Cantelli)

how large s s.t. with high confidence (**P**robably)
know that Σ_{best} is accurate (**A**pproximately **C**orrect)?

Once Over, Lightly

Pick *loss* or (missmatch) *penalty* function $\mathcal{L}_{\Sigma,\xi}[u(\cdot), y(\cdot)]$

stationarity assumption:
training data $(u_i(\cdot), y_i(\cdot))$ as well as test pairs $(u(\cdot), y(\cdot))$ distributed according to same (unknown) prob measure P

min total loss $\sum_{i=1}^{s} \mathcal{L}_{\Sigma,\xi}[u_i(\cdot), y_i(\cdot)] \rightsquigarrow \Sigma_{\text{best}}$

good model := expected error on test pairs is ≈ 0
goal: show Prob (training data led to good model) ≈ 1

if set of possible inputs too rich, hopeless
discrete-time: finite length inputs
here continuous-time: match data to $O(T^k)$ on $[0, T]$

main result: for c.t. nets of dim $= n$, and $\sigma = \tanh$, enough

$$s = O(n^6 + n^3 \log k)$$

i.e. $O(\log k)$ for fixed dimension

Precise Formulation

Assume $\sigma = \tanh$, $\dim = n$ (and for simplicity, $m{=}p{=}1$)

consider output jet $Y_{\Sigma,\xi}(\mu) := (y(0), y'(0), \ldots, y^{(k)}(0))$

corresp to $\mathcal{C}^{k-1}$ input with $\mu = (u(0), u'(0), \ldots, u^{(k-1)}(0))$

$$\mathcal{L}_{\Sigma,\xi}[\mu,\eta] := \frac{\| Y_{\Sigma,\xi}(\mu) - \eta \|^2}{1 + \| Y_{\Sigma,\xi}(\mu) - \eta \|^2}$$

write $\zeta = (\mu, \eta) \in \mathcal{Z} := \mathbb{R}^{k+(k+1)}$

$\forall$ sample vector $Z_s = (\zeta_1, \ldots, \zeta_s) \in \mathcal{Z}^s$: empirical loss

$$\widehat{\mathcal{L}}_{\Sigma,\xi}[Z_s] := \left(\frac{1}{s}\right) \sum_{i=1}^{s} \mathcal{L}_{\Sigma,\xi}[\zeta_i]$$

$\forall$ Borel probability measure P on $\mathcal{Z}$:

$$\mathrm{E}^P \mathcal{L}_{\Sigma,\xi} := \text{ expectation of } \mathcal{L}_{\Sigma,\xi}[\cdot]$$

Sample Complexity Estimate

Theorem. For all small $\varepsilon, \delta > 0$,

$$s \;=\; \Omega\left(\left(n^6 + n^3 \log k\right) \frac{\log(1/\varepsilon)}{\varepsilon^2} \;+\; \frac{\log(1/\delta)}{\varepsilon^2}\right)$$

$\Rightarrow$ with confidence $P^s(Z_s) > 1 - \delta$:

$$\mathrm{E}^P \mathcal{L}_{\Sigma,\xi} \;<\; \widehat{\mathcal{L}}_{\Sigma,\xi}[Z_s] + \varepsilon \quad \forall\, \Sigma, \xi \;(\dim = n)\; \forall\, P$$

i.e.: inf loss $= \lambda \Rightarrow$ with prob $>1-\delta$, expected error $<\lambda+\varepsilon$
e.g.: $\lambda = 0$ if fit experimental data perfectly

note: even $\exists s < \infty$ not obvious
(cf. Macintyre-EDS STOC'93: based on Wilkie, o-minimality)

estimate by Warren/Khovanski (via Macintyre-Karpinski)
and Pollard's pseudodimension (via Haussler)

related: min $\widehat{\mathcal{L}}_{\Sigma,\xi}[Z_s]$? EDS, Adv Comp Math'96:
$\rightsquigarrow$ differential-topological study of critical points of $\widehat{\mathcal{L}}_{\Sigma,\xi}[Z_s]$

Universal Models for Analog Computing

"computing" by continuous-space devices = ?

$\exists$ universal models, like Turing Machines for digital?

quantify resources (e.g. computation time); ask e.g.:

P $\subsetneq$ AP ("analog polynomial time")?   NP $\subseteq$ AP?

"devices":

$$x(t+1) \;=\; f(x(t),u(t)) \quad y(t) = h(x(t))$$

$$x(t) \in \mathbb{R}^n \quad u(t) \in \mathbb{R}^m \quad y(t) \in \mathbb{R}^p \quad x(0) = \xi$$

will assume f, h very regular
also: finite-dimensional, deterministic
(pde: many pathologies due to nonregularity
Pour-El, Richards, et al; wave eqn, etc)
also well-known difficulties if nonunique sols
–c.f. work of logicians on computing by ode's

Possible Formulations of I/O

One possibility: input data $\sim$ initial state ξ
assume guaranteed $x(t) \to$ equilibrium as $t \to \infty$
Then let "answer" := $x\,(+\infty)$
done in e.g. Hopfield-type nets

Not totally clear how to formulate
resource scaling with input "size"
unless # dimensions scaled too

More natural idea, in comparison to digital:
look at values of output variable $y(\cdot)$
and ask how long they take to be produced
as compared to the time it takes to read input

compare w/digital comp: decision problems
(standard way to benchmark computing power)
so use binary scalar inputs $u(t)$
— need suitable notion of "binary" outputs $y(t)$

Input/Output Encodings

encode binary words $\alpha = \alpha_1 \ldots \alpha_n \in \{-1, 1\}^+$ as inputs

$$u_\alpha : \mathbb{N} \to \mathbb{R} \quad (\alpha_1, \ldots, \alpha_n, 0, 0, 0, \ldots)$$

$x_\alpha(\cdot) :=$ state traj with this u_α and $x(0) = \xi$
$y_\alpha(\cdot) :=$ output traj $h(x_\alpha(\cdot))$

Reading outputs: **don't** want to depend on
infinite precision "external observer"
e.g. "ans is NO if $y(t_0) = 0$ and YES otherwise"

well-behaved system if "unambiguous" output:
$\exists T = T_\alpha$ s.t. $-\frac{1}{2} \leq y_\alpha(t) \leq \frac{1}{2}\ \forall\, t < T$, $|y_\alpha(T)| > 1$
interpret as "accept" if > 1, "reject" if < -1

wb-system Γ computes in polynomial time if

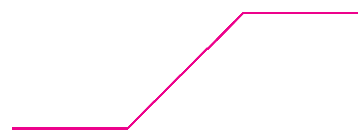
$$T_\alpha \leq cn^k \quad \forall\, \alpha \in \{-1, 1\}^n$$

Let $L_\Gamma := \{\alpha \mid \Gamma \text{ accepts } \alpha\}$

Dynamical Nets as Universal Models

Let AP = analog polynomial time
= class of languages L_Σ for Σ computing in polytime,
where Σ : $x^+ = \vec{\boldsymbol{\sigma}}(Ax + Bu)\,,\ y = Cx$,

$$\sigma = \begin{cases} -1 & \text{if } x < -1 \\ x & \text{if } -1 \leq x \leq 1 \\ 1 & \text{if } x > 1 \end{cases}$$



Theorem. (Siegelmann-EDS'92/94; *Science* 4/95)

$$\Gamma \ \textbf{“reasonable”} \Rightarrow L_\Gamma \in \textbf{AP}$$

“reasonable” includes bounded & continuous
piecewise combs of elementary funcs, real constants
(precise def: can compute bits using sparse oracle)
not “reasonable”: discontinuities in transitions
– (Boolean input) Blum-Shub-Smale not included (open!)

Also: P $\subsetneq$ AP ; NP $\not\subset$ AP (unless PH=Σ_2)

(via AP = P/poly = Turing/advice + Karp-Lipton)

Summary

- introduced dynamic (recurrent) nets
- characterized internal state diagnosis (observability)
- form determined by function (identifiability)
- minimal if observable
- characterized transitivity of action (reachability)
- approximation (density)
- estimated complexity noisy identification / PAC Learning
- nets are universal model class for analog computing

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* Fourier Recurrent Networks *

discrete-time, $\sigma(x) = e^{ix}$

closed-form algorithm for realization (noise-free case)

Assume $m = 1$ and generic conditions:

- $b_1, \ldots, b_n$ nonzero
- $\sigma(b_1), \ldots, \sigma(b_n)$ distinct
- $c_1, \ldots, c_n$ nonzero
- observability: $\ker C \cap A^{-1}(\mathbb{Z}^n) = \{0\}$
- column space of

$$\begin{bmatrix} \sigma(b_1) - 1 & \cdots & \sigma(b_n) - 1 \\ \sigma(2b_1) - 1 & \cdots & \sigma(2b_n) - 1 \\ \vdots & & \vdots \\ \sigma((n+1)b_1) - 1 & \cdots & \sigma((n+1)b_n) - 1 \end{bmatrix}$$

has no nonzero lattice points ($\bigcap \mathbb{Z}^{n+1} = \{0\}$)

* `Procedure I/O` $\rightsquigarrow (A,B,C)$ *

Apply $2n$ different integer inputs of length 1:

$$\begin{aligned}
& C\vec{\boldsymbol{\sigma}}(Ax_0), \\
& C\vec{\boldsymbol{\sigma}}(Ax_0+B), \\
& \vdots \\
& C\vec{\boldsymbol{\sigma}}(Ax_0+(2n-1)B) \\
= \;& \tilde{C}\tilde{A}^0\tilde{B}, \tilde{C}\tilde{A}\tilde{B}, \ldots, \tilde{C}\tilde{A}^{2n-1}\tilde{B}
\end{aligned}$$

where

$$\tilde{C} = [C_1\sigma(A_1x_0), \ldots, C_n\sigma(A_nx_0)]$$

$$\tilde{A} = \begin{bmatrix} \sigma(b_1) & & \\ & \ddots & \\ & & \sigma(b_n) \end{bmatrix}, \quad \tilde{B} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$

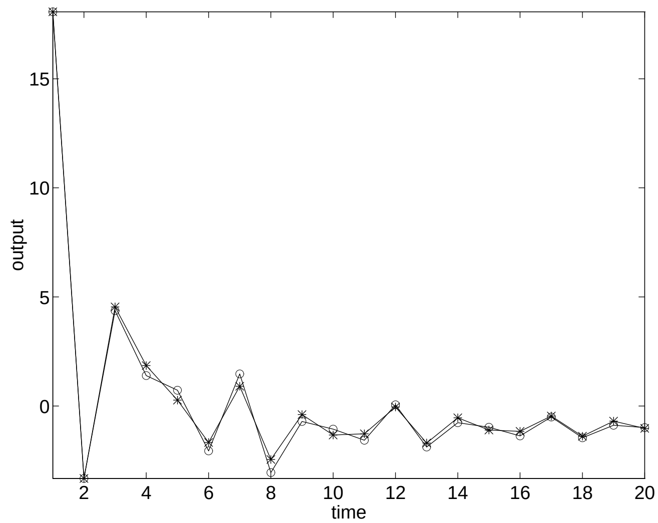
now linear realization to recover eigenvalues of $\tilde{A}$:

$$\sigma(b_1), \sigma(b_2), \ldots, \sigma(b_n)$$

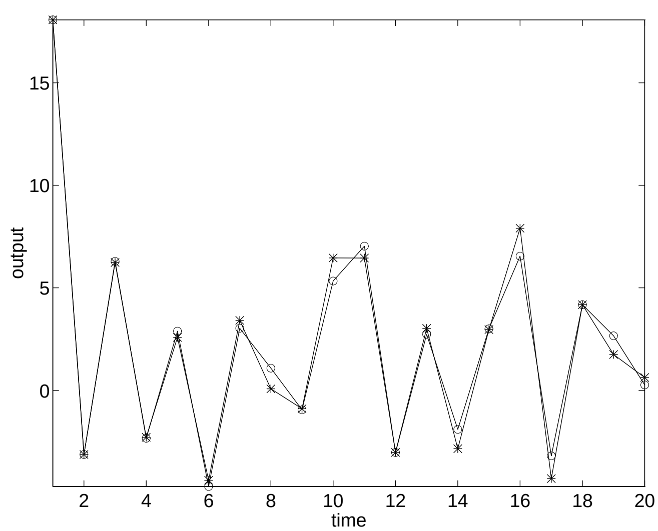
$\rightsquigarrow B$ (but we can't "invert" σ, so need a bit more)
next find C and A; also Cx_0 and $x(1)$ for any control

* MATLAB Implementation; check w/noise *

dim 4 example



input sequence of 20 ones; noise 10^{-3}
(true (∘) and appx (*) outputs)



random input sequence, length 20; noise 10^{-3}